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Project Title:	Mechanosensing and Mechanotransduction by the Na-H exchanger NHE1		
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	In single-cell organisms plasma membrane Na-H exchangers (NHEs) function as osmoregulators. In multicellular organisms, we propose that the functions of NHEs have evolved to include mechanosensing and mechanotransduction as an adaptive response to the specialized needs of mechanically integrated tissues. One component of this adaptive response is that in multicellular organisms NHEs are structurally linked to the actin-based cytoskeleton. In metazoans, NHEs catalyze an electroneutral exchange of extracellular Na+ and intracellular H+, which regulates intracellular pH and volume homeostasis, and they act as plasma membrane anchors of the actin cytoskeleton, which regulates cells shape and adhesion. Conserved responses of metazoan cells to gravity-generated forces include changes in cell shape, cytoskeleton organization, and cell adhesion. The objective of this proposal is determine the role of the uniquitously expressed Na-H exchanger NHE1 in gravity-generated responses by investigating the hypothesis that NHE1 is a critical		

Task Description:	determinant in the sensing and transducing of mechanical strain. Because mechanical forces are important regulators of connective tissue homeostasis, and fibroblasts sense force-induced deformations (strains) in the extracellular matrix we will use fibroblast models developed in our lab that express mutant NHE1 with impaired ion translocation or cytoskeletal anchoring. In Aim 1, the role of NHE1 in mechanosensing will be investigated by studying NHE1 activity in whole cells and in excised membrane patches in response to mechanical loading, by determining whether cytoskeletal anchoring by NHE1, cell adhesion, and integrin activation are necessary for NHE1 activation by mechanical loading. In Aim 2, the function of NHE1 in mechanotransduction will be investigated by using conventional microscopy and atomic force microscopy to determine whether NHE1 activity or cytoskeletal anchoring are necessary for dynamic changes in cell topology, and in remodeling of the actin cytoskeleton and cell-substrate adhesions in response to mechanical loading. Processes controlling cytoskeleton remodeling, including de-novo actin polymerization and the activation of Rho family GTPases will be determined in response to mechanical loading, and the dependence of NHE1 in this response will be evaluated. These studies explore a previously unrecognized role of NHE1 in mechanosening and mechanotransduction, and they are designed to further our understanding of how gravity-generated forces regulate cell functions. The overall objective is to identify molecular components of a mechanotransduction unit in fibroblasts. In contrast to the wealth of information known about how cells sense chemical stimuli, little is known about how cells sense mechanical forces and transduce this information into biochemical events and cell responses. An understanding of this process will likely provide signficant insight into how mechanical forces modulate cell and tissue function, morphogenesis, differentiation, and remodeling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We identified the ubiquitously expressed Na-H exchanger NHE1 as a necessary component of a mechanotransduction unit in fibroblasts. This finding is significant in understanding how cells sense and respond to mechanical force, which is a important for determining how mechanical force drives normal processes such as development, tissue organization, wound-healing, and cell proliferation and migration, and contributes to pathological conditions such as atherosclerosis and fibrosis.
Task Progress:	The study of mechanotransduction is an exciting new direction for our research program. We previously developed most of the cell models described in our research plan, and have considerable expertise in studying cytoskeletal dynamics and activities of Rho family GTPases in response to growth factors and hormones. The first year of our NASA award was spent developing a uniaxial loading device with NASA Ames researchers Dr. Eduardo Almeida and Nancy Searby, and optimizing conditions for using these devices to determine responses to cell stretch by using biochemical assays and live-cell imaging. New data obtained since our previous progress report include the following: Aim 1. Our previous report included data showing that mechanical force increases intracellular pH and that this is blocked by pharmacological agents inhibiting activity of the Na-H exchanger, NHE1. We have now confirmed that mechanical force increases NHE1 activity and phosphorylation. NHE1 activity was determined in a nominally HCO3--free Hepes-buffered medium as the rate of pHi recovery (dpHi/dt) from an NH4Cl-induced acid load. Compared with unloaded cells, a 5 min 1% stretch and release induced a marked increase in NHE1 activity. With mechanical loading, NHE1 activity became quiescent at a higher pHi compared with unloaded cells, suggesting a change in the set-point value. We currently are determining the kinetics (Vmax, Km) of NHE1 activation with mechanical load. Chemical signals (serum, platelet-derived growth factor [PDGF], epidermal growth factor [EGF], thrombin, and lysophosphatidic acid) increase phosphorylation of NHE1 at C-terminal serine residues. Phosphorylation of NHE1 is necessary for increased NHE1 activity by growth factors and hormones, but not by osmotic stress. We found that NHE1 phosphorylation increases with mechanical load. Cells labeled with 32P-othophosphate were not loaded or loaded for 5 min with 1% stretch and release, NHE1 was immunoprecipitated with NHE1 antibodies generated in our lab, and its phosphorylation determined by autoradiography. Silver staining before autoradiography confirmed relatively equal gel loading. A future objective is to determine whether phosphorylation of NHE1 is necessary for its activation by mechanical force. We currently are using mass spectrometry to identify amino acids phosphorylated by mechanical loading and then will ask whether alanine substitutions of phosphorylated residues impairs mechanical activation of NHE1. Although activation of NHE1 by mechanical force has previously been reported, to our knowledge these data are the first to show that mechanical force induces an increase in NHE1 phosphorylation. Aim 2. We next asked whether NHE1 is necessary for mechanotransduction by investigating established responses to mechanical force. Actin dynamics Reorganization of actin filaments and an increase in actin polymerization are conserved responses to mechanical force. We found that mechanical load induces a dramatic increase in the bundling of actin filaments (stress fibers) in CCL39 fibroblasts. We currently are asking whether NHE1 is necessary for this response by testing actin bunding in the presence of NHE1 inhibitors and in cells lacking NHE1. We also observe an increase in the abundance of F-actin (increased polymerization) in response to mechanical load, by using an in-gel assay of Triton X-insoluble proteins. The abundance of F-actin was similar in quiescent NHE1-expressing CCL39 cells and in NHE1-deficient PS120 cells, which are derived from a parental CCL39 clone. With mechanical loading, however, there was an almost 2-fold increase in F-actin in CCL39 cells, but no increase in PS120 cells. To determine whether the impaired F-actin response in PS120 cells is due to lack of NHE1, we measured actin free barbed end formation in response to mechanical load in PS120 cells and in PS120 cells stably expressing wild-type NHE1 (WT cells). We chose to investigate free barbed end formation because it is an index of filament servering and occurs prior to de novo actin assembly. By using an assay with biotin actin to label actin free barbed ends, we found that the number of free barbed ends was similar in quiescent WT and PS120 cells. With mechanical loading free barbed end formation increased in both cell types, however, the response in PS120 cells was attenuated compared with WT cells. Fluorescence microscopy was used to quantify actin free barbed ends, which indicated newly formed free barbed ends were uniform around the cell at the submembrane region. A key regulator of de-novo actin assembly is Arp2/3, a seven-subunit protein complex that builds crosslinked filament arrays by nucleating new actin filaments from the sides of pre-existing filaments. Although Arp2/3 activity is controlled by interaction with nucleation promoting factors (NPFs) of the WASP- and Scar/WAVE-family, in a separate study we

	recently found that phosphorylation of the Arp2 subunit is a previously unrecognized mechanism necessary for activity of the Arp2/3 complex. Our studies indicate that if Arp2 is not phosphorylated, Arp2/3 cannot be activated by NPFs. We found that Arp2 undergoes a regulated phosphorylation in mammalian cells labeled with 32P-orthophosphate and treated with EGF, and in response to mechanical load. Pretreating CCL39 cells with the NHE1 inhibitor EIPA attenuated Arp2 phosphorylation with loading, suggesting the response is partially NHE1-dependent. Focal Adhesion Dynamics. We first asked whether mechanical loading regulates the assembly of focal adhesions. Before mechanical loading, CCL39 cells had few, small focal complexes, as determined by immunolabeling with antibodies to phosphotyrosine. After a 5 min 1% load and release there was a marked increase in focal adhesion assembly, with larger and more abundant adhesion complexes. We found that activity of focal adhesion kinase (FAK) increased with mechanical loading in CCL39 fibroblasts, as determined with Y397 phosphospecific antibodies, but not in NHE1-deficient PS120 cells that are derived from a parental CCL39 clone. The response was mostly restored in PS120 cells stably expressing wild-type NHE1 (WT cells). The abundance of pFAK in quiescent CCL39, PS120, and WT cells was similar. Hence, NHE1 is not necessary for steady-state FAK activity, but is necessary for increased FAK activity with mechanical loading. We also found that recruitment of FAK to focal complexes was induced by mechanical loading of CCL39 cells, but not PS120 cells. Immunolabeling fixed cells for total FAK indicated a diffuse cytosolic distribution of FAK in both cell types. With mechanical loading immunolabeling detected in punctuate peripheral aggregates in CCL39 cells was attenuated in PS120 cells. These data suggest that NHE1 functions in mechanotransduction because it is in-part necessary for conserved responses to mechanical load, including for de novo F-actin assembly and the activation of FAK its recruitment to focal complexes and adhesions. We currently are preparing a manuscript describing these data in the context of being a first report showing that NHE1 plays a central role in mechanotransduction. Additionally, these findings were used as preliminary data in a grant proposal on mechanotransduction that was submitted to the National Institutes of Health. We recently learned that the proposal received a fundable score and grant support will begin in January 2007.
Bibliography Type:	Description: (Last Updated: 07/31/2007)
Abstracts for Journals and Proceedings	Shubov A, LeClaire L, Madson N, Almeida E, Barber DL. "Mechanotransduction in Fibroblasts is Mediated by the Na-H exchanger NHE1." 45th Annual Meeting of the American Society for Cell Biology, San Francisco, CA, December 10-14, 2005. 45th Annual Meeting of the American Society for Cell Biology, Abstracts, December 2005. , Dec-2005